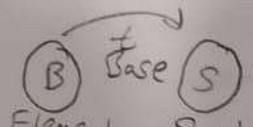
NUMBER SYSTEMS



$b \in B$ $\quad a_i \in S$

$a_i \in \{0, 1, 2, \dots, t-1\}$

Fixed Point Number System

Floating Point Number System

1) $b = S(a_n t^n + \dots + a_1 t + a_0 + a_{-1} t^{-1} + \dots + a_{-\ell} t^{-\ell})$

2) $b = S \sum_{i=-\ell}^{n} a_i t^i$

3) $b = S\, a_n a_{n-1} \dots a_1 a_0 , a_{-1} \dots a_{-\ell}$

$t=2$ Binary Number System $a_i \in \{0, 1\}$

$t=8$ Octal Number System $a_i \in \{0, 1, 2, 3, 4, 5, 6, 7\}$

$t=10$ Decimal Number System $a_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$t=16$ Hexadecimal Number System $a_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F\}$

$b_t = S\, a_n a_{n-1} \dots a_0 , a_{-1} \dots a_{-\ell}$

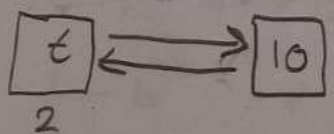
MSB, LSB, MSB, LSB

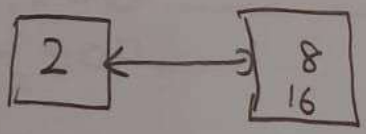
S: sign

$a_n a_{n-1} \dots a_0$: integer

$a_{-1} \dots a_{-\ell}$: fractional

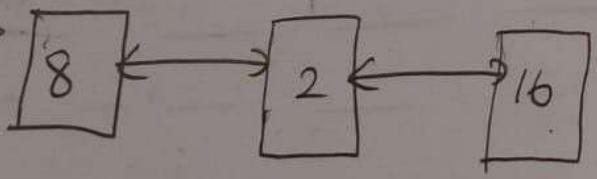
BASE CONVERSION





$2^3 = 8$

$2^4 = 16$



$t \rightarrow 10$

$b = (1\,1\,0\,1\,0\,,\,1\,0\,1\,1\,0)_2 \rightarrow$

$b = 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^1 + 1 \cdot 2^{-3} + 1 \cdot 2^{-4}$

$b = (26,6875)_{10}$

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1 n → t (2, 8, 16)

Öğ: $(353)_{10} = (101100001)_2$

b = N, M

LSB ↓ MSB

MSB ↓ LSB

A	t
	B
G	

A = B·T + G

353	a_i
176	1 → a_0 (LSB)
88	0
44	0
22	0
11	1
5	1
2	0
1	0
0	1 → a_8 (MSB)

Fractional Conversion

$r - P = \epsilon^{-9}$

$a = P \cdot \frac{\ln r}{\ln t} \Rightarrow 9 = P \cdot \frac{\ln 10}{\ln 2} \cong 3{,}32P$

$M = (0{,}65625)_{10} = (\text{-----})_2$

$9 = 3{,}32 \cdot 5 = 16{,}6 \cong 17$

a_{-i}	P_i	t
MSB —	0,65625	2
1	0,31250	
0	0,6250	
1	0,25	
0	0,50	
LSB 1	0,0000	

Öğ: $(100/11/010) = (472)_8$

$(1001/1010) = (9A)_{16}$

Öğ: $(BA92)_{16} = (\quad)_2$

$(1011\ 1010\ 1001\ 0010)_2$

The Complementation of the Numbers

- Complementation to $t-1$ (Radix minus one notation)
- Complementation to t (Radix notation)

- Complementation to $(t-1)$: Radix minus one notation

$T = t^{n} - t^{-m} - N$

$N = \overset{\wedge}{12,79}$ $\quad T = 10^{2} - 10^{-2} - 12,79 = 87,20$

$N = 101 \longrightarrow T = 010$

$(654)_{10} \Rightarrow T = (345)_{10}$

$$\begin{array}{rr} & N \;\; 12,79 \\ + & T \;\; 87,20 \\ \hline & 99,99 \end{array}$$

- The Complemat tot

$T_t = t^{n} - N = T_{t-1} + 1\,(LSB)$

$N = 1279 \Rightarrow T_t = 10^{4} - 1279 = 8721$

$T_t = T_{t-1} + 1\,(LSB) = 8720 + 1$

$= 8721$

$N\,(101)_2 \Rightarrow T_1 = 010 \longrightarrow 2^{3} - 2^{0} - (101)_2$

$T_2 = 011$

$8 - 1 - 5 = 2$

$$\begin{array}{rr} & N \;\; 101 \\ + & T \;\; 010 \\ \hline & 111 \end{array}$$

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NEGATIVE NUMBER

- Sign and Magnitude
- Diminished - Radix Notation (Radix minus one)
- Radix Notation

	S+M	t-1	t
+5	00101	00101	00101
-5	10101	11010	11011
+9	01001	01001	01001
-9	11001	10110	10111

Fixed Point Arithmetic

- Radix Notation (t Base)
- Radix minus one Notation (t-1 Base)

Case 1 A>0 B>0
case 2 A>0 B<0 (vice versa)
Case 3 A<0 B<0

Radix Notation
case A>0 B>0

$$A \quad 0\ a_{n-2}\ a_{n-3} \cdots a_1\ a_0, a_{-1} \cdots a_{-k}$$
$$B \quad 0\ b_{n-2}\ b_{n-3} \cdots b_1\ b_0, b_{-1} \cdots b_{-k}$$
$$S_S\ S_{n-2}\ S_{n-3} \cdots S_1\ S_0, S_{-1} \cdots S_{-k}$$

$S_S = 0 \quad A+B < t^{n-1}$

$S_S \neq 0 \quad A+B > t^{n-1}$ (overflow)

OR:

$$\begin{array}{r} 13 \\ +11 \\ \hline 24 \end{array}$$

$$\begin{array}{r} 01101 \\ +01011 \\ \hline 11000 \end{array}$$

overflow

→ 8 bit yer ayrılmış olsaydı

$$\begin{array}{r} 0000\ 1101 \\ +0000\ 1011 \\ \hline 0001\ 1000 \end{array}$$

Case $A<0$, $B<0$

$$\begin{array}{r} -A \\ +\ -B \\ \hline -(A+B) \end{array}$$

$(\epsilon-1)\ a_{n-2} \cdots a_1\ a_0 \mid a_{-1} \cdots a_{-k}$

$(\epsilon-1)\ b_{n-2} \cdots b_1\ b_0 \mid b_{-1} \cdots b_{-k}$

$$\begin{array}{rr} A & -4 \\ +\ B & -7 \\ \hline & -11 \end{array} \qquad \begin{array}{r} 11100 \\ +\ 11001 \\ \hline \cancel{1}\,01001 \end{array}$$

$$\begin{array}{r} \epsilon^{\wedge}-A \\ +\ \epsilon^{\wedge}-B \\ \hline 2\epsilon^{\wedge}-(A+B) \end{array}$$

$\Downarrow$

$\epsilon^{\wedge} + [\epsilon^{\wedge} - (A+B)]$

$1 + [\epsilon^{\wedge} - (A+B)]$

$\downarrow$ ignore $\quad \epsilon^{\wedge}$ digit

Diminished Radix Arithmetic ($\epsilon-1$ Base)

- $A>0$, $B>0$

$$\begin{array}{r} +A \\ +\ +B \\ \hline +(A+B) \end{array}$$

Ör: $\begin{array}{r} +4 \\ +\ +11 \\ \hline +15 \end{array} \quad \begin{array}{r} 00100 \\ +\ 01011 \\ \hline 01111 \end{array}$

Ör: $\begin{array}{r} +6 \\ +\ +11 \\ \hline +17 \end{array} \quad \begin{array}{r} 00110 \\ +\ 01011 \\ \hline 10001 \end{array}$

$\Uparrow$ overflow

- $A<0$, $B<0$

$$\begin{array}{r} -A \\ +\ -B \\ \hline -(A+B) \end{array} \qquad \begin{array}{r} \epsilon^{\wedge}-\epsilon^{-k}-A \\ +\ \epsilon^{\wedge}-\epsilon^{-k}-B \\ \hline 2\epsilon^{\wedge}-2\epsilon^{-k}-(A+B) \end{array}$$

$\Downarrow$

$\epsilon^{\wedge}-\epsilon^{-k}-(A+B) \quad \epsilon^{\wedge}-\epsilon^{-1} + [\epsilon^{\wedge}-\epsilon^{-k}-(A+B)]$

uymuyor

MSB $\qquad$ LSB

End - Around - Carry

Ör: $\begin{array}{r} -3 \\ -11 \\ \hline -14 \end{array} \quad \begin{array}{r} 11100 \\ +\ 10100 \\ \hline \textcircled{1}\,10000 \\ +\ \quad\quad 1 \\ \hline 10001 \end{array}$

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• A > 0 , B < 0

$$\begin{array}{r} A \\ +\ -B \\ \hline \end{array} \qquad \begin{array}{r} A \\ +\ t^{n} - t^{k} - B \\ \hline t^{n} - t^{-k} - (B - A) \end{array}$$

ÖR: −3 → 11

$$\begin{array}{r} 11100 \\ +\ 01011 \\ \hline 100111 \\ +\ \quad\quad 1 \\ \hline 01000 \end{array}$$

ÖR: −11 → 3

$$\begin{array}{r} 10100 \\ +\ 00011 \\ \hline 10111 \end{array}$$

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